

JAMES CENAWOOD

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EDUCATION

Cornell University, College of Engineering

August 2024 – May 2028

Bachelor of Science in Computer Science

GPA: 3.9

Completed: Probability & Statistical Inference, Stochastic Processes, Data Science, Discrete Math, Linear Algebra, OOP and Data Structures, Functional Programming, Calculus

In Progress: Machine Learning, Analysis of Algorithms

EXPERIENCE

Northwell Health | Data Science Intern

June 2026 – August 2026

- Architected and built a NL2SQL data exploration agent for the hospital's EPIC data warehouse, retrieving across ~40,000 schema documentation files and drafting schema-grounded SQL and analytical insights
- Crafted a gold-standard benchmark bucketed by question type to isolate vocabulary gaps that deeper ranking cannot close; maintained a fail-closed design, with no patient-data access and deterministic validation before any SQL is returned
- Won (tied for first) the intern Kaggle competition with a CatBoost/TabNet blend, Optuna tuning, and grouped cross-validation to prevent group leakage across folds

Cornell Physical Intelligence | Software Subteam Lead

December 2025 – Present

- Lead an 8-person software subteam building the autonomy stack for Anduril's AI Grand Prix (mapless visual navigation, collective-thrust/body-rate control); the team's entry qualified for Virtual Qualifier 1
- Built the AI-native development environment behind the team's iteration loop; PowerShell automation driving a closed-source Windows simulator end to end (launch, menu navigation, focus retention, official scoring, run summarization), plus cascading context management and a reusable hypothesis-loop encoding qualification-mode telemetry restrictions as hard constraints
- Wrote and administered 20+ technical interviews using an original drone-themed problem set, adapting standard algorithmic questions to the domain the team actually works in

Cornell Extended Reality | Software Team Member

August 2025 – Present

- Designed a persistent-memory backend for an AI smart-glasses assistant; typed, multi-tenant store for conversation episodes, time-scoped facts, summaries, and a per-person relationship graph supplying profile context to a dialog LLM
- Built a natural-language person resolver mapping descriptions to stored identities via multi-pass name matching, emitting disambiguation hints (visual descriptor, affiliation, hobby) for ambiguous references

PROJECTS

Surpassing the Leader: Death Game Solver

github.com/paldrdr/SURPASSING-THE-LEADER

- Formalized the penultimate two-person death gamble from the manga [Usogui](#) into a finite, simultaneous-move, zero-sum stochastic game and produced a certified exact solution; authored the accompanying paper
- Proved a per-player quotient that collapsed 5.27×10^9 states to 289,374,121 classes; a potential function sorts these into 1,201 layers solvable in one backward sweep without search
- Gated every stored value behind an explicit equilibrium certificate within 10^{-6} of the full 60×60 stage matrix; implemented the sweep in Rust and Python with byte-for-byte agreement on all shared regions
- Reduced an estimated 3.7–5.1 year computation projection to a 5,432-second twelve-core session; certified root value gives the first Dropper a 0.5449 win probability

AlgoSplit: Full-Stack Algorithmic Hypertrophy App

algo-split.vercel.app

- Built a full-stack training app predicting weekly stimulus across 29 muscle regions before a split is run; increased average net weekly stimulus 35% across 30+ splits for initial users
- Formalized the hypertrophy stimulus-fatigue literature into a weekly model, with an exercise parser resolving lift names into movement pattern, target muscles, resistance profile, and axial fatigue contribution
- Shipped web/iOS (FastAPI, Supabase, React, Expo Go) with authentication, workout logging, history, 3D stimulus visualization, and multi-split comparison; rewrote the analysis kernel in Rust for 335% faster rendering behind a Python parity gate

Sentinel: Intelligent Waiting-Room Triage | 1st Place, URMIC × WICC × Gemini Hackathon

sentinel.thekoppe.com

- Designed and built a Python engine converting a live waiting-room triage stream into five bounded patient risk scores with hard-override rules for critical cases, replacing a single assessment taken at intake
- Built a queue controller with anti-flap controls (promotion hysteresis, demotion persistence) that kept patient ordering stable under noisy input, plus board-level metrics like high-risk count and acuity drift

TECHNICAL SKILLS

Languages: Python (NumPy, Polars, pandas, scikit-learn, PyTorch, OpenCV), OCaml, TypeScript, SQL, Rust

Tools: Git, FastAPI, LangGraph, Supabase, Vercel, SQLite/FTS, BigQuery, Hydra, Optuna, Weights & Biases, Docker, L^AT_EX, Neovim

Interests: Tournament Poker (top-3 Jane Street & Citadel sponsored), Game Design, VR, 3D Animation